

Treasures from the Year of Light



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Lighting has come a long way, from flint stones and firewood to fluorescent lamps and now light emitting diodes (LEDs). Today, we continue to see innumerable developments in this segment, right from space and power saving to smartness and smart uses (to make other objects smart). As we draw to the end of the year, which the United Nations designated as International Year of Light and Light Based Technologies (www.light2015.org/Home), let us take a look at some such technologies and products.

Light for the new-age farmer

Royal Philips, a company that has been into horticultural lighting for several years now, has recently opened GrowWise City Farming Research Centre in the Netherlands for developing tailor-made LED light growth recipes that will help farmers to grow tasty, high-yielding crops indoors, all round the year. Their research will first focus on leafy vegetables, strawberries and herbs, and later branch out to find ways to grow more carbohydrate-rich crops like wheat and potatoes indoors.

The large facility is a clean and sterile environment that completely blocks off natural light and air to exercise full control over the growing conditions. It has four-layered mechanised planting racks in each of its eight climate rooms, which makes a total growing surface of 234sqm. Each plantation layer is equipped with connected, fully-customisable LEDs, including blue, red and far-

red LEDs that are designed and formulated specifically for growing crops. Since LEDs are highly energy-efficient and produce less heat, these are much cooler and can be placed closer to the plants and positioned optimally for uniform illumination.

This research of Philips is very crucial for future sustainability as more and more farming land is being engulfed by the expanding cities. Growing food indoors at optimal conditions with minimal effort and resources, and the ability to grow foods closer to home, will encourage more people to take up farming, too.

Controlling the functioning of humans

One of the key focus areas of this year's Strategies in Light Conference held at Las Vegas, USA, was human-centric lighting projects, designed to optimise or improve some aspects of human behaviour. Extensive research is being done around the world on how a human body responds to light exposure and how this can be used to improve the mental or physical state of a person.

The Seattle Mariners' stadium, a major league baseball stadium in Seattle, USA, for example, has fitted the home team locker room with new solid-state lighting that has features to tweak the players' circadian rhythm for optimum alertness. One of the speakers has described methods for integrating sensors into lighting systems to produce light optimised for a given activity. The non-invasive sensors automatically categorise occupants' activities and adjust the lights for performance improvement—of the person, not just the light!

Philips SchoolVision is a similar solution that helps students to focus on their work, facilitates teachers to teach better and also results in energy savings for the school. According to the time of the day and activity planned, teachers can choose between four modes that mimic the natural patterns of daylight that humans respond to. During a test, for example, Focus mode sets the light intensity to highest and the colour tone to cool, supporting focus and concen-

Philips' new GrowWise indoor farm will revolutionise food production (Image courtesy: www.lighting.philips.com)

